

Graduation proposal 6/18/2026

StudentDetails

Name	Stijlaart, Casey C.
Student number	4280989
Student upn	l466488@fhict.nl
Study mode	Fulltime
Location	Eindhoven

Main choices

HboiLayer	Hardware Interfacing
Motivation for choosing this HBO-I layer	
ItAreaName	Embedded Systems
Continue personalized program?	No
Preferred language during internship	English

OrganisationDetails

Type of company/institution	FontysInternal
Name	Mindlabs
Visiting address	Locomotiefboulevard 101
Zipcode	5041SE
City	Tilburg
Country	The Netherlands
Phone	
Website	https://www.mind-labs.eu/over-ons/

CompanyMentor

First name + Family name	Arjan Groeneweg
Department	Interaction and UX designer/developer
Position	Coordinator and Senior Lecturer UX & Interaction Design
Background (highest education)	Master's Degree, Media Innovation
Background URL	https://www.linkedin.com/in/arjangroeneweg/?locale=nl
Phone	
Email	a.groeneweg@fontys.nl

InternshipDuration

Choose duration of internship
Choose startmoment of internship

20 weeks
Week 1

OnStage assignment

OnStage assignment
If yes, link to assignment in OnStage

No

ProjectDescription

Title

Virtual Human & Holobox

1. Problem/opportunity analysis

Describe the company and the context they operate in. Who are the different stakeholders involved? Are there related projects? What problem(s) or opportunities are relevant for your assignment, and why? Describe the starting situation. Who is affected by the problems? Etc.

MindLabs is a creative technology studio based in Tilburg that develops immersive, interactive experiences at the intersection of digital and physical worlds. One of their current projects is the Holobox — a holographic display installation in which a Virtual Human interacts with visitors, primarily children aged 8–12. The stakeholders involved are MindLabs as the developing organisation, the Experience Design intern responsible for the Virtual Human's appearance and behaviour, and the end users: children and potential future visitors interacting with the installation. For the Virtual Human to feel truly responsive and natural, it needs to know where a person is in the space — not just horizontally, but also their approximate height, so it can adapt its behaviour accordingly (for example, kneeling when interacting with a child). Currently no sensing system exists for this. A camera-based solution is an obvious approach but raises privacy concerns in a context involving children, and offloads significant processing to external systems.

There is therefore an opportunity to design a dedicated embedded sensing system using non-camera sensors that can determine a person's 3D position in real time, in a privacy-preserving way, and output that data reliably to the Holobox pipeline. The system also needs to be designed with future extensibility in mind, as scenarios involving multiple people in the space are anticipated.

2. Assignment

Specify initial assignment, goals, scope, products to be delivered/realized, technology to be used.

The goal of this internship is to research, design, and build an embedded multi-modal sensing system capable of estimating the real-time 3D position — including height — of a person within the Holobox interaction space, without the use of cameras.

The assignment includes:

– Researching and comparing suitable non-camera sensing technologies (e.g. mmWave radar, ultrasonic, IR, UWB) on criteria such as spatial resolution, height discrimination, embedded feasibility, and robustness in a real environment with children

- ?Selecting and combining two or more complementary sensors and designing a fusion architecture
- ?Implementing a processing pipeline in C/C++ and/or Python on an embedded platform (MCU or Linux SBC) that outputs a reliable position and height-category estimate in real time
- ?Exploring the use of lightweight on-device machine learning for height classification or person selection
- ?Designing the system output interface so it can be consumed by the Unreal Engine pipeline
- ?Documenting the architecture, data flow, and design decisions

Deliverables include a working embedded prototype, a technical research report covering sensor selection and fusion approach, implemented interaction logic (position + height output), and a system architecture document. The system will be designed with multi-person extensibility as a future-facing requirement, though single-person tracking is the focus of this internship.

3. Relation to research

Which areas of the assignment are expected to require a research attitude? What could be related research questions?

Several areas of this assignment require a genuine research attitude rather than straightforward implementation. The primary research challenge is sensor selection and fusion: there is no single established non-camera solution for room-scale 3D human localisation on an embedded platform, and the trade-offs between sensor types in this specific context (small space, children, reflective surfaces, real-time requirement) need to be systematically investigated through literature study and practical experimentation.

A second research area is on-device processing: determining where in the pipeline computation should live — on a microcontroller, a Linux SBC, or split between them — requires evaluation of latency, power, and accuracy trade-offs that depend on the chosen sensors and algorithms.

A third area is the application of lightweight ML for height classification on a constrained device, which requires researching appropriate model architectures, training data strategies, and deployment methods for edge inference.

Possible research questions include:

- ?Which combination of non-camera sensors provides sufficient 3D spatial resolution to distinguish adult and child height categories within a room-scale interactive space?
- ?How can sensor fusion be implemented on an embedded platform to produce robust, low-latency position estimates under real-world noise conditions?
- ?How can a lightweight ML model be trained and deployed on an edge device to classify person height from fused sensor data with acceptable accuracy?
- ?How should the system architecture be designed to support single-person tracking now while remaining extensible to multi-person scenarios?

4. Guidance

What kind of support does the company provide on personal and IT skills related to your domain? How will you be mentored? What background (Bachelor/Master/Phd) does your company mentor have and what

expertise is available in the company? How many hours of guidance per week?

The primary company mentor is Arjan Groeneweg, an interaction and UX designer/developer at MindLabs. He will provide guidance on how the sensing system fits into the broader Holobox experience, the interaction requirements it needs to meet, and professional development within the company context. Mentoring sessions are expected approximately once per week.

For technical guidance specific to embedded systems, sensor integration, and software architecture, additional team members with relevant technical backgrounds are available within MindLabs. This ensures that both the interaction design context and the embedded engineering depth of the project are adequately supported.

5. Personal development goals

Which aspects on personal or professional development will you further develop? Select a realistic three to five aspects to work on. This can be aspects that you are already good at and you want to improve or skills that are not yet on an acceptable minimal level. Document how your plans fit in your assignment and how you will work on it in more detail. Use previous received feedback/ feedforward from others (e.g. semestercoach/mentor as input).

#	Goal	AchievementDescription
1	Deepen practical ML on edge skills beyond prototyping level	By week 12, design, train, and successfully deploy a lightweight classification model on the embedded target platform, evaluated against a defined accuracy threshold on collected test data.
2	Develop system architecture design skills for multi-component embedded systems	By week 8, produce a complete and reviewed system architecture document covering sensor fusion data flow, processing distribution, and output interface – validated through feedback from a technical mentor.
3	Strengthen research attitude: problem into structured research	By week 4, formulate a set of concrete, answerable research questions derived from literature study, and maintain a structured research log translating an openthroughout the project.
4	Improve cross-disciplinary collaboration in a professional creative tech	Throughout the internship, conduct at least bi-weekly alignment sessions with the Experience Design intern and Arjan to ensure the technical system meets interaction design requirements, reflected on in progress reports.

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6. (Optional) other important remarks